PITCHMIND

Football Knowledge Base

Tactical concepts, coaching frameworks and evidence-aware analysis

A curated reference for understanding football phases, team movement, formations, player roles, goalkeepers, set pieces, match states, metrics and AI-supported explanations.

Evidence first. Conditional tactical explanations. No invented match facts.

Prepared from PitchMind knowledge documentation, a curated catalogue of 45 linked sources, and the pitch-regions-18-v1 tactical convention.

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Football domain knowledge atlas

Purpose and source discipline

This atlas gives PitchMind a broader association-football domain reference for explaining verified match evidence. It combines a curated set of public laws, governing-body coaching and competition analyses, and peer-reviewed research. It is not an exhaustive encyclopedia or a transcription of source pages. The linked sources are paraphrased, and their examples remain source examples rather than facts about a PitchMind match.

The named-coach profile files are analytical lenses built from general tactical concepts. They do not claim that PitchMind authors or agents worked with the named coaches, reproduce their private methods, or imitate their voices. No profile, article, formation convention or model output can supply a missing event, player attribute, instruction, intention or match fact.

Evidence hierarchy for explanations

1. **Canonical match facts:** validated PitchMind event and tracking records, preserving source IDs, order, provenance, clock and period semantics.
2. **Derived observations:** versioned calculations from validated facts, with formulas, coordinate conventions, coverage and missing-data behavior.
3. **Classifications:** outputs from documented, evaluated detectors with known error bounds.
4. **Tactical interpretations:** conditional explanations that point back to observations and name plausible alternatives.
5. **General knowledge:** this atlas and its sources; it can explain a mechanism but cannot establish that mechanism occurred in the selected match.

When evidence is insufficient, reduce specificity or abstain. Separate association from causation, action from intent, and source examples from the selected match.

Laws, match format and competition context

The IFAB Laws of the Game are the authority for association-football laws. Check the edition applicable to the match and any permitted competition modifications. For analysis, Laws 3, 5, 7, 8, 10, 11, 12, 13, 14 and 15 are frequent context for substitutions, referee authority, time allowance, restarts, goals, offside, fouls/cards, free kicks, penalties and throw-ins. The analyst should record what the feed reports and should not adjudicate a referee decision from incomplete event data.

Clock representation is provider-specific. Preserve period, source clock, added-time notation and elapsed-time conversion as distinct fields. A displayed `90+N` does not establish how much time remains. A foul late in a match may interrupt a transition, concede a dangerous restart, stop play, or have another context; “tactical foul” is an interpretation requiring sequence evidence, not a law-defined event label.

State the code and format. Eleven-a-side association football is not interchangeable with futsal, beach soccer, small-sided development formats or adapted youth football. Player count, pitch size, offside application, substitution rules and restart procedures can differ. Do not transfer a formation or metric across codes without a documented mapping.

The game as interacting phases and scales

Use the phase taxonomy as context rather than as a complete tactical explanation. FIFA's World Cup analysis describes in-possession phases (including build-up, progression, final-third play and counterattack), out-of-possession phases (press/block, counterpress, recovery and defensive transition), and set-play phases (FIFA phases of play). Analyze the **team**, the relevant **unit**, and the **player** interaction together (FIFA scales of the game).

For each possession sequence, ask:

- What phase is supported by the source, and what event or possession change starts/ends it?
- How are width, depth, line spacing, support, cover and access to the next line represented in data?
- What does the opponent do in response, and which response remains a plausible alternative?
- Does the observed action change territory, create an option, produce a shot/restart, retain possession or lose the ball?
- What is not observable: scan, body shape, communication, assignment, intent, pressure quality or instruction?

In possession

Common principles include creating length and width, separating opponents, positioning between/around lines, supporting the ball, progressing through/around/over pressure, switching play, combining with a third player, carrying to commit a defender, and occupying the box and rebound areas. These are options, not a mandatory sequence. A safe recycle may be better supported than a forced forward pass when the lane, timing or rest-defence evidence is absent.

UEFA's competition observations show how context changes the solution: the 2022/23 Champions League report discussed a trend toward playing short from goal kicks, while the 2023/24 technical observers highlighted compact defending and space-oriented marking, and UEFA's 2024/25 reports cover wide-area control, goal-kick strategy and competition-specific patterns (2022/23 report overview, 2023/24 report overview, 2024/25 technical report catalogue). These are season- and competition-bound observations; do not treat an observed trend as a universal prescription.

Out of possession

Describe a defensive organization through line height, width, depth, compactness, cover, pressure, screening, marking reference and the space the unit is willing to concede. A high press is a collective relation among the first presser, cover, screened passing routes and the rest of the team; one player chasing the ball does not establish a team press. A mid/low block can steer play wide, protect central access, delay an attack or prepare to counter. Each choice exposes something: space behind a high line, central gaps if a midfielder jumps, weak-side access if the ball-side unit over-shifts, or reduced forward outlet if the block retreats.

FIFA's tactical material describes compactness, coordinated shifting, pressure and cover as connected but distinct principles (FIFA out-of-possession organization). UEFA's technical observers have also described teams defending space rather than tracking opponents everywhere; marking references must be observed rather than inferred from a shape label.

Transitions and rest defence

After a verified regain, distinguish an immediate counter from a secure first action and a later settled attack. Check forward lane, ball carrier, runners, support, opponent recovery and second-ball structure. After a verified loss, inspect the closest pressure, cover behind it, central-screening, far-side narrowing and recovery routes. A counterpress can recover the ball or delay progress; if bypassed, it can leave open space behind. FIFA's core-principles series treats third-player combinations, switches, high pressing, counterattacks and counterpressing as separate mechanisms (FIFA core principles).

Rest defence is the attacking team's protection against a counter while it has the ball. Describe the actual players and spaces behind/beside the attack, not merely possession percentage or a formation name. Consider the trade-off between committing runners to threaten the box and retaining cover for a turnover.

Formations, roles and matchups

A formation is a lineup-derived starting shorthand. A phase shape is a time- and possession-dependent arrangement. Keep the two distinct. A nominal 4-3-3 may defend in a 4-1-4-1 or attack in a back-three/five-lane structure, but these labels require lineup plus positional evidence and a stated counting convention.

Evaluate matchups by mapping the actual numerical and spatial relations: first line against pressers, midfield access, wide channel, half-space, depth behind the last line, and cover around the ball. For each proposed overload, specify area, time, involved players and nearby cover. Then state the counter-risk. An apparent 3v2 may be neutralized by a cover shadow or a defender stepping out; a wide 2v1 may be less useful if the carrier cannot face forward or the far-side defender can cover. A matchup suggests questions to test; it does not guarantee advantage.

Roles are requirements that can be shared across different player types, not assumptions about individuals. A pivot may need to offer a passing angle, scan, receive under pressure, protect central transitions and vary distribution. An overlapping full-back may need timing, speed, crossing or cutback choices, recovery and a complementary wide player who can move inside or hold width. A winger may stretch depth, isolate a defender, combine or recover into a block. A pressing forward may curve the run, screen a pivot, coordinate triggers and remain available after a regain. A target striker may connect direct play and occupy defenders, but needs support for second balls and protection if isolated. Validate these attributes from approved player evidence; never infer them from a role label or reputation.

For substitutions, compare like-for-like windows and phases before/after the recorded change. Report the player's observed role and team behavior; score, opponent changes, dismissals, restart mix and sample size are competing explanations. A temporal change after an entry does not prove the substitute caused it or reveal the coach's intention.

Goalkeeper as part of team structure

Analyze the goalkeeper in possession as a possible extra passing option: starting position, available angles, pressure, first touch, pass range, target and support for the next action. A short goal-kick build-up can create an extra line but can also increase turnover risk near goal; a direct route can bypass pressure but requires a target, runner and second-ball support. UEFA's season reports show that teams use different goal-kick strategies, not that one is always best.

Out of possession, assess starting depth relative to defensive line and ball, ability to cover space behind, position for crosses/cutbacks, and one-on-one decisions. These assessments need aligned goalkeeper, ball, attacker and defender positions and the shot/cross context. One frame may show a position; it does not establish a trend, anticipation skill or decision quality. FIFA's goalkeeper resources cover build-up and transition positioning as distinct contexts (build-up, transition positioning).

Restarts and second phases

Treat corners, free kicks, penalties, throw-ins and goal kicks as distinct phases with their own laws, player assignments and temporal boundaries. For each restart, capture delivery, first movement, blockers/screens where visible, target zone, first contact, clearance, edge occupation, rebound/second ball and transition protection. Separate a designed routine hypothesis from an observed movement pattern; repeated motion alone does not reveal an instruction.

For defending, identify whether the observable organization appears zonal, player-oriented or mixed, then inspect contest, keeper traffic, near/far-post protection, edge cover and counter outlet. Do not mark the assignment of a player unless the evidence shows it. A clearance is not the end of the phase if possession remains contested. FIFA's corner analysis includes examples of freeing an edge shooter, blocker/runner movements and zonal/player-oriented organization; these are examples to compare, not templates (FIFA set-play analysis).

Late game, dismissals and unusual states

Condition interpretation on score, time semantics, players on the field, restart type and accumulated evidence. A narrow lead or need to score may coincide with deeper defending, more direct play, altered width, additional box presence or fewer players behind the ball. These are tactical possibilities, not facts about intent. A dismissal changes numerical availability and may prompt role redistribution, but whether the team changes formation or risk posture must be observed. Last-minute fouls require law context plus transition, location, player, restart and card evidence. Avoid unsupported claims about time-wasting, deliberate injury, fatigue or referee intent.

Player development, practice and format effects

Football performance is not only technical or tactical. The FA's Four Corner Model treats technical/tactical, physical, psychological and social development as connected lenses and cautions coaches to adapt support to the individual (The FA Four Corner Model). This is a player-development framework, not a match-state classifier. Age, maturation, sex, competition level, training history and format matter when transferring findings.

Small-sided/conditioned games change the information and action demands by manipulating player numbers, area per player, pitch geometry, targets, touch constraints, rules and work/rest pattern. Research reviews report that such changes can alter technical actions and collective tactical behavior, but effects are design- and sample-dependent; a practice task does not establish transfer to a full match (Bezerra et al., systematic review, Sarmento et al., systematic review). The model should not recommend a training prescription from match data without age, workload, health and coaching context.

Match analysis, metrics and causal limits

Football analysis must define actions and outcomes operationally, record sample/population and include match situation. A review of match-analysis research found variation in outcome variables and emphasized standardized definitions and situational/interactional context (Sarmento et al., match-analysis review). At minimum, metrics should identify formula, unit, denominator, time/phase window, player/team scope, provider/version, event coverage, missingness, calibration and permitted comparisons.

- Counts depend on exposure; use a justified denominator before comparing teams or players.
- Per-90 values do not control for possession, role, phase, opposition or event coverage.
- Passing completion does not measure progression, difficulty or decision quality by itself.
- Shot volume, goals and saves do not alone establish chance quality or goalkeeper performance.
- Distance and high-speed running do not diagnose fatigue or intent.
- A formation/press label requires an operational definition and validated observation method.
- Before/after comparisons show association. Causal claims need an appropriate design and controls.

AI-based tactical analysis can map collective movement, shape, networks and patterns, but input modality, tracking coverage, sample selection, competition, sex and age constrain generalization. A recent systematic review of AI methods in football tactical/collective-dynamics research describes a diverse method landscape and explicit inclusion boundaries (AI and collective dynamics review). PitchMind should state when its analysis is event-only versus tracking/video-assisted and must not imply that an event feed observes body orientation, off-ball movement or collective shape.

Knowledge-base article record

The machine-readable reviewed article catalogue records topic, code, age/competition context, source type, publication date where known, review date, a short paraphrase and use/rights guidance. Before adding a source, verify the direct URL, publisher, content type, context and date. Prefer the law-making authority for rules, governing-body technical reports for their competitions, original research for empirical claims and transparent systematic reviews for broad evidence. Record limitations and do not treat a later or more popular article as automatically stronger evidence.

Football tactical knowledge base v1

Purpose and boundaries

This knowledge base gives PitchMind's explanation model a consistent football vocabulary and a disciplined way to interpret evidence. It is organized around repeatable coaching concepts, not personal impersonations. The owner-provided list of coaches spans different eras, competitions and tactical approaches; this document does not claim personal experience working with them, attribute a specific view to each person, or imitate anyone's voice. It synthesizes common coaching and analysis concepts into evidence-checkable cards.

The concise, machine-readable cards in `football-tactical-knowledge-base.v1.json` are embedded in AgentWorker. Each of the six specialist profiles plus the evidence-first default references only the relevant cards. The selected profile's principles and cards are placed in one bounded model request. The model still gets a selected-time evidence packet, returns an untrusted candidate, and passes through the existing deterministic verification boundary.

At runtime, the model receives a bounded `promptSummary` for the selected cards; the detailed mechanisms, evidence requirements, analysis steps and limitations remain in the versioned knowledge-base resource and this full guide. This resource explains concepts; it is not a source of match facts, player facts, validated metrics, referee decisions, training data, or an approved tactical detector. A phrase in the knowledge base cannot make a missing observation true.

A common analysis method

Analyze football at three connected levels:

1. **Team:** phase, organization, spacing, field occupation and response to the opponent.
2. **Unit:** a line or local group working around the ball, including numerical relations such as 2v1 or 3v2.
3. **Player:** an observed action and the player's relationship to teammates, opponents, the ball and direction of play.

Move from broad context to a local mechanism and then to an evidenced outcome. FIFA's coaching analysis framework uses team, unit and player scales and illustrates how roles and numerical relationships change during an attack; PitchMind uses that as a vocabulary aid, not as evidence about any PitchMind match (FIFA: Scales of the game).

For each explanation, follow this chain:

`match/time/revision → phase → observed positions/events → tactical mechanism → opponent response → outcome → uncertainty`

- Pin all statements to the packet's `matchId`, `canonicalRevision`, projection revision, selected time and sequence watermark.
- Separate **observed** source facts from **derived** measurements, **detector classifications**, and **interpretation**.
- Use actual evidence IDs for factual claims. Use later evidence only if it is within the selected-time packet.
- Prefer a small, precise claim over a complete-sounding story. If two mechanisms fit, name the ambiguity.
- Keep counterfactuals conditional: “If the wide defender steps to the ball, the pass may release the runner” is a tactical option, not proof that this happened or would have succeeded.

Shared vocabulary, metric definitions and evidence confidence

Use one explicit vocabulary across articles, APIs, detectors and model prompts. Similar terms from different feeds must not be treated as equivalent until their definitions are compared. Distinguish possession from phase, pressure from a press, a local overload from a whole-team numerical advantage, progression from penetration, and a recorded event from an inferred tactical mechanism. FIFA's football-language work describes a unified analysis vocabulary and notes that provider definitions and calculations can differ (FIFA: The Football Language).

Every metric entry should define its name and purpose, numerator, denominator, unit, aggregation window, population/player set, provider, version, missing-data policy and known limitations. Keep totals, per-match rates, per-90 rates, per-ball-in-play rates, completion rates, proportions, averages, medians, percentiles and standard deviations distinct. A rate is not comparable if its denominator, phase, role, competition, minutes or source coverage differs (FIFA: Data analysis definitions).

For each explanation, separate **observed source fact**, **derived measurement**, **detector classification**, and **tactical interpretation**. Name evidence IDs/time windows for factual claims. State relevant coverage, sample size and calibration; provide alternatives when evidence supports them. Use confidence tied to evidence quality, not model fluency. Abstain or use less specific wording when missingness, clock semantics, mapping or coordinate calibration can change the conclusion. A source article supports general football concepts; it never supports a claim about the selected match.

Reusable metric record

For each metric document: `metricId`, definition, formula, unit, denominator, team/player scope, phase and time window, source/provider, version, minimum coverage, missingness behavior, intended comparisons and prohibited interpretations. For example, “pressures per 30 minutes out of possession” requires a validated pressure-event definition and out-of-possession denominator. It cannot be substituted with raw pressure count or interpreted as pressure quality without outcomes and context.

Scenario and counterexample record

For each model-evaluable concept, maintain paired, fictionalized scenarios with (1) evidence that satisfies the operational definition, (2) a near-miss that looks similar but lacks one required element, (3) missing or conflicting data, (4) the bounded explanation expected in each case, (5) evidence citations and temporal cutoff, and (6) a reviewer note on plausible alternatives. Store scenario evidence separately from canonical match data and label synthetic examples. Include counterexamples such as a lateral pass mislabeled as a switch, pressure mislabeled as a team press, a high shot count mislabeled as good chance quality, and distance mislabeled as fatigue. A polished answer must not score better than a correct abstention.

Concept	Supporting scenario	Near-miss/counterexample	Expected bounded response
In-game adaptation	Same phase repeated: opponent changes the press; team repeatedly changes build-up role; changed exit is observed.	Shape differs after a substitution or score change, but no opponent response/adjustment sequence is documented.	Describe the observed sequence and alternatives; do not assign coaching intent or causal success.
Defensive line	Multiple aligned defenders step as the ball travels; runner, cover and keeper are visible.	One defender steps in a single frame, or a later offside flag appears without relevant positions.	State line action only when measured; do not call an offside trap from the later event alone.
Scanning/receiving	Video coding records a head turn before receipt and the subsequent touch/action.	Pass completion exists but video/scan/body orientation is unavailable.	Describe scan only in the first case; otherwise report the pass/action and mark perception unknown.
Chance context	Shot event plus valid location, pressure, goalkeeper/defender positions and a calibrated model result.	Goal or miss exists without model features, or shot count is used as quality.	Separate shot and result; withhold chance-quality or xG conclusion without validated inputs.

1. Movement dynamics across the field

Phases to identify

Classify the best-supported phase before naming a tactic:

- **In-possession organization:** build-up, progression through/around/over a unit, creation in the final third, box occupation, restart attack.
- **Out-of-possession organization:** high press, mid-block, low block, retreat/recovery, defending a restart.
- **Transition to attack:** immediate action after a verified regain; direct counter or retained/circulated possession.
- **Transition to defend:** immediate action after a verified loss; counterpress, delay, retreat, or recovery shape.

Possession, phase and shape are not interchangeable. A team may keep the ball without progressing, press without using a high block, or occupy a low block while remaining compact. FIFA's out-of-possession analysis describes a block's width, vertical spacing, collective shifting and pressure as separate principles (FIFA: Team organisation out of possession).

Movement relationships to inspect

When tracking is valid and sufficiently complete, examine:

- **Width and depth:** lateral spread and threat behind/through the opponent. Keep measured team spread distinct from a player's nominal position.
- **Team length and line distances:** space between defensive, midfield and forward units. A team can narrow horizontally, compress vertically, or do both.
- **Centroid and ball-relative location:** where the observed group is centered relative to the ball and pitch. A centroid alone does not reveal shape or intent.
- **Ball-side shift and weak side:** whether multiple players move across together and what space opens away from the ball. A single player moving toward the ball is not a collective shift.
- **Between-lines occupation:** a player or unit positioned in a gap between opponent units. This requires time-aligned opponent and receiver locations, not only a successful pass.
- **Overload and isolation:** a local numerical or positional advantage in a defined area. State the count, participants, area and time.
- **Rest defence:** attacking-team players positioned to protect against a counter while teammates attack. Identify the players/space behind the ball; possession share is not evidence of rest defence.
- **Recovery runs and transition protection:** observed movement to delay the counter, recover a line, protect central access or regain the ball.
- **Opponent response:** the opponent may shift, drop, engage, hold, cover, press, retreat, or expose a lane. Report only responses visible in source evidence.

Spatial measurement requirements

Before making a spatial or movement claim, validate all of the following:

1. Pitch dimensions and the source's coordinate units.
2. Coordinate origin, axes, orientation and team-relative attack direction for each period.
3. Player and team identity mappings, goalkeeper/outfielder treatment, and active roster at the selected time.

4. Frame timestamps, event clock alignment, sample cadence and missingness.
5. The definition/version of any derived zone, line, compactness, overload or distance threshold.

If any item is unknown, step down in specificity. Say “the sequence progressed into the wide area” only if the event classification supports that; do not say “the winger moved 12 metres into the left half-space” from uncalibrated coordinates. A snapshot can describe a snapshot. A movement trend needs multiple aligned observations. Sparse frames cannot establish acceleration or velocity.

How to describe dynamics clearly

Use a compact sequence:

> “As [observed ball/event], [identified unit/players] [observed movement]. That created/closed [space or option] according to [derived evidence]. The opponent [observed response]. The next evidenced action was [outcome]. [Limit].”

Do not fill bracketed parts from football convention when the packet is silent.

2. Substitutions and the late-match impact question

What a substitution does and does not prove

A canonical substitution proves that an identified player left and another entered at a recorded time, provided the source event and roster mapping are valid. It does **not** prove why the change was made, whether the outgoing player was tired or injured, what instruction was given, or whether the formation changed. Official match laws define the substitution procedure; tactical purpose remains a separate question (IFAB Law 3: The Players).

Tactical mechanisms to consider as hypotheses

A change in personnel can potentially alter:

- **Role and lane coverage:** a like-for-like role, a player taking another line/zone, or different cover of an opponent.
- **Width and depth:** a runner stretching the back line, a wide outlet, a player between lines or an additional box target.
- **Pressing and recovery:** pressure on the first pass, cover behind the presser, sprint/recovery capacity, or the ability to hold a block. Do not infer fitness from name/age or substitute status.
- **Progression and retention:** carrying, passing range, aerial outlet, ability to receive under pressure or protect possession. Require player/event evidence; a profile or reputation is insufficient.
- **Transition balance:** more forward runners can add threat while reducing protection behind the ball; both sides of the trade-off require evidence.
- **Set-piece roles and defensive matchups:** only if the relevant restart and assignments are observed.
- **Game-state risk:** a team may attack more directly, retain the ball, protect a lead or seek a goal, but motive is not established by the substitution itself.

A defensible pre/post assessment

1. **Pin the change:** outgoing/incoming identities, team, period, clock semantics, event ID and active roster before/after.

2. **Set the observation window:** for example, comparable possession phases or a bounded elapsed-time window before and after. Explain why the window is appropriate. Stop at selected time.
3. **Describe role evidence:** validated role/position data or tracking shows where the incoming player actually operated. Without this, report only the personnel change.
4. **Compare the same measures:** same definitions and coverage before/after for team width/length, line spacing, regain location, progression, pressing sequences or shot/entry outcomes.
5. **Control context:** score, opponent adjustment, dismissal, other substitutions, restart-heavy periods, possession share and sample size can all explain a change.
6. **State what happened, not why:** “After the substitution, the team used more evidenced forward passes in this window” is an association. “The substitute caused the improvement” requires a separate causal design and cannot be inferred from a short before/after sequence.

For a “last-minute substitution,” first define the match clock and period. Stoppage time is part of the period's allowance and provider feeds may encode it differently; never assume a label such as `90+4` means four minutes remain. IFAB Law 7 describes allowance for lost time, including substitutions and disciplinary sanctions (IFAB Law 7: Duration of the match).

Safe explanation pattern

> “The substitution is recorded at [validated time]. In the observed window afterward, [measured team behavior] changed/did not change. This is consistent with [bounded tactical possibility], but the packet does not establish the instruction or cause.”

3. Attacking 2v1 scenarios

Definition and context

A 2v1 is a local, time-specific relationship between two attacking players and one relevant defender in a defined corridor or zone. State whether the ball carrier is included in the two. Check whether another defender is close enough to cover; the local picture can be 2v1 in a lane while the broader defense still has cover. FIFA coaching material illustrates creating and exploiting 2v1s through overlaps, underlaps, movement into pockets and using a spare player (FIFA: Developing wide attacking patterns, FIFA: Creating and exploiting overloads). These are coaching examples, not proof of an event in a PitchMind match.

Attacker's decision tree

- **Carry at the defender:** may commit the defender and release the teammate, or may let the defender delay and protect depth. The model can describe the option only when carrier, defender and teammate positions are observed.
- **Pass to the free player:** depends on lane, timing, weight, receiver orientation and the defender's commitment. A completed pass does not prove that the lane was open earlier.
- **Use a third player:** a wall pass, set, or third-player run can move the ball around the defender. The event sequence must include the intermediate player and the third action.
- **Overlap:** a teammate runs outside/beyond the ball carrier; this can stretch or pin a defender and create a crossing/carrying lane.
- **Underlap:** a teammate runs inside the carrier, often toward an inside channel; direction and destination need tracking or event evidence.
- **Recycle or protect:** if the defender holds shape and cover closes the direct option, the evidence may support retention or return rather than forced penetration.

Defender's decision tree

The defender may engage the carrier, delay, protect depth, screen a pass, show the attack toward a sideline, or wait for covering support. Determine what happened from position/event evidence. A defender who delays may have prevented penetration even if the attackers retain possession; a defender who engages may expose a pass but can also disrupt the attack. Do not label either “wrong” from outcome alone.

Analysis checklist

- Identify the corridor, timestamp, ball, attackers, primary defender and covering defenders.
- Confirm team/player mappings, validated coordinates, attack direction and frame cadence.
- Describe the initial numerical relation and whether it changes as players move.
- Record the ball carrier's action, defender response, support movement and next action.
- State the outcome—progression, cross, shot, retained possession, turnover, foul or unknown—only when present.
- Keep the tactical trade-off and alternative response visible; avoid retrospective certainty such as “he should have passed.”

Without validated tracking or a canonical local-overload detector, event-only data cannot prove a 2v1, defender commitment, open passing lane, overlap/underlap, or available option.

4. Last-minute fouls and tactical-foul analysis

Separate match fact from tactical interpretation

Represent these as distinct facts or stages:

1. The source's foul/physical-offence event and its ID/time.
2. Whether play stopped or advantage was applied, as recorded by an authoritative event source.
3. Restart type and location, if validated.
4. Card/caution/dismissal event, if present.
5. Subsequent restart, attack, shot or other outcome.
6. Tactical interpretation: whether the foul interrupted a transition, protected a zone, conceded a restart or may have managed tempo.

IFAB Law 12 governs fouls and misconduct, while Law 7 governs allowance for time lost. These laws are the authority for rules; the analytics model should not adjudicate referee correctness (IFAB Law 12: Fouls and misconduct, IFAB Law 7).

When “tactical foul” may be used

Use this as a cautious interpretation only if evidence shows the defending team was facing an opponent transition and the foul interrupted that progression. Record the phase, location, participants and restart/card aftermath if known. Even then, use “the foul stopped/interrupted the observed transition” rather than “the coach ordered a tactical foul.” A late foul is not automatically cynical, deliberate time-wasting, instructed or strategically successful.

Last-minute timing

- “Last minute” needs an explicit definition: final 60 seconds of regulation, final 60 seconds of the period including added time, or another competition-specific window.
- Preserve provider time and period labels. Do not convert `45+N` or `90+N` into wall-clock remaining time without validated feed semantics.
- Score, card state, restart position and time all influence the possible tactical consequences, but must be pinned to the selected-time packet.
- Explain actual consequences: free kick/penalty/restart, card, dismissal, attack stopped, continued advantage or lost time. Do not calculate unvalidated danger or claim the event changed the result.
- Never advise, praise or normalize a player deliberately fouling an opponent. Describe the recorded event and its football consequences neutrally.

Safe explanation pattern

> “At the source-recorded time, [team/player if verified] committed a foul during [supported phase]. Play restarted with [recorded decision]; [card/next action] is/is not in the selected evidence. This interrupted [observed progression] if the event sequence establishes it. Intent and referee correctness are not determined here.”

5. Pressing and defensive movement

Use the chain **trigger** → **pressure** → **cover/support** → **opponent response** → **regain or escape** → **next action**. A possible trigger (backward pass, poor control, wide reception, receiver facing their own goal) is a hypothesis until the ordered events and/or tracking show it. A trap requires an evidenced route constraint and opponent response; a touchline reception alone is not a trap.

For collective shape, inspect horizontal compactness, vertical distances between lines, ball-side shift, space behind the last line and access through the center. These are separate measurements. A low block can be compact; a high line can be stretched. Pressure by one player is not a collective press. A regain after pressure is temporal sequence, not proof of causation.

FIFA's out-of-possession coaching material describes narrowing horizontally, compressing vertically, moving across as a unit, keeping shape and applying pressure as related but distinct ideas (FIFA: Team organisation out of possession). Require validated tracking and coverage before using those descriptions for PitchMind data.

Pressing triggers and routes out

Potential triggers include a backward or square pass, a receiver's poor or bouncing touch, a pass toward the touchline, an opponent receiving with a closed body orientation, a slow aerial ball or an isolated receiver. These are candidate cues, not universal rules. Name a trigger only if event/tracking evidence records it. Analyse the full chain: cue, first presser's angle and speed, cover shadow or screened lane, supporting players stepping, opponent exits, response and whether the press regained or merely delayed play. One sprint toward the ball is pressure, not proof of a coordinated press. A trap requires evidence that exits were constrained and the opponent responded.

When describing how a team beat pressure, consider playing between pressers, around the press into a free player, over it to a target/space, using a bounce or third-player action, carrying through a vacated lane, switching to the weak side, or retaining/recycling until a safe route appears. FIFA describes passing through/around pressure and high-press organization, traps and risks; these examples inform vocabulary but do not prove PitchMind match events (FIFA: Passing the press, FIFA: High press).

State the trade-off for either side: bypassing a line can create an advantage if possession is retained but expose a central turnover; pressing high can shorten the route to goal but expose space behind the back line if pressure or cover is bypassed. A successful pass alone does not prove the receiving lane was open before the pass, and a regain after pressure does not prove pressure caused it.

6. Transitions as a separate analysis layer

Treat transition as a short, distinct interval following a validated regain or loss, with a declared start event and end condition. Preserve source event ID, sequence and period/clock semantics. Analyse initial reaction, nearest pressure, cover/screen, forward or safe outlet, opponent recovery, next possession phase and outcome. FIFA distinguishes the in-possession and out-of-possession phases and treats the reaction immediately after loss/regain as tactically significant (FIFA: Defensive transitions, FIFA: Core principles of play).

After a regain, distinguish an immediate counterattack from secure possession followed by an organized attack. A counterattack claim needs evidence of rapid forward exploitation while the opponent is disorganized; a long pass or later goal alone is insufficient. After a loss, distinguish counterpressing from delaying the ball and recovering shape. If first pressure is bypassed, inspect recovery runs and protection of central access. Do not infer an instruction or intent from the sequence. Compare alternatives and state if the feed is too sparse to identify the transition.

7. Chance creation and prevention patterns

Describe attacking patterns as observable combinations rather than labels inferred from outcomes. Candidate patterns include third-player combinations, wall passes, overlaps, underlaps, switches of play, runs behind a high line, rotations to create a free player, cutbacks, crosses, central combinations and attacks on second balls. Record the sequence, players and lanes only when identity/event data or calibrated tracking supports them. A third-player combination requires an evidenced intermediate action; a switch requires a change of side; a run behind requires a time-aligned starting line and movement; a cutback requires a backward/diagonal delivery from the byline area under a documented definition. FIFA's core-principles series covers third-player combinations and switching play among its in-possession topics (FIFA: Core principles of play).

Pair each pattern with possible defensive responses: screen the central lane, pass runners between defenders, engage or delay the carrier, protect depth, shift the block, cover the far-post/cutback zone, or contest first/second contact. These are options, not prescribed assignments. Assess the actual defending sequence and outcome. A shot or goal does not by itself establish that a pattern was well executed; a blocked or failed attempt does not prove the decision was poor. State whether evidence describes the pattern, a generated chance, or only the final event.

8. Physical and tactical context

Physical measures can describe demands occurring within a tactical phase, but should be contextualized by role, phase, possession state, time on pitch, match period and source coverage. Keep external measures (distance, speed bands, accelerations/decelerations) separate from internal response measures (such as heart-rate or perceived-load data), with units and collection methods explicit. Compare only compatible definitions and samples; normalize denominators where appropriate and preserve raw values and denominators. FIFA contextualized physical metrics using phase-of-play information and notes limitations of measurement methods, particularly for explosive, dynamic actions (FIFA: Physical-analysis background and method, FIFA: Monitoring training load).

Do not infer fatigue, fitness, effort, motivation, injury, decision quality or tactical intent from sprint count, distance, substitution, age or a single time window. A high running total may reflect role and game state; a low count may reflect possession, coverage, minutes, tracking gaps or role. Present physical and tactical observations side by side and use cautious association language. Physiological or training-load data requires its own validated source, privacy controls and specialist review.

9. In-game adaptation and opponent response

Separate a team's broad playing principles from match-specific choices and from an observed in-game adjustment. Use the sequence **initial approach** → **opponent response** → **observed adjustment** → **new space/constraint** → **second response** → **next outcome**. Compare like phases and account for score, personnel, dismissal, restart, field zone and sample size. A tactical change may address one exposed space while opening another; an apparent response could also reflect a phase or personnel change. Do not infer coaching intent solely from changed positions or possession share. FIFA describes the consistency/adaptability trade-off and illustrates tactical adjustments during a match (FIFA: Consistent vs adaptable approaches, FIFA: Japan v Sweden, adapting attacking strategies mid-game). These sources inform concepts only; do not transfer their team-specific findings to PitchMind matches.

10. Defensive line management

Describe the defensive line as a time-varying unit: height relative to the goal, distance to the midfield/forward lines, movement when the ball travels, depth coverage, and relationship to the goalkeeper. Distinguish stepping up to compress space from dropping to protect depth. A coordinated step may compress a between-lines space or attempt to catch runners offside, but mistiming can expose a run behind; dropping may protect depth while creating space in front. These are conditional trade-offs, not quality verdicts. FIFA defines defensive line height and team length as separate metrics (FIFA: Defensive line height and team length).

For a line movement claim, require time-aligned locations for the relevant defenders, ball, attackers/runners and goalkeeper, calibrated pitch direction, period/time and adequate frame coverage. Offside status requires the law-defined positions at the relevant instant and suitable authoritative event/geometry; do not infer a successful offside trap from an opponent being flagged later. For a ball in behind, inspect passer pressure, line movement, runner start/timing, cover defender, keeper depth and actual reception/shot sequence. A through pass alone does not prove the line was high, disorganized or deliberately beaten (FIFA: Balls in behind the defence).

11. Scanning and receiving under pressure

Scanning is a player's observable head movement away from the ball to gather information; a scan count does not reveal what information was perceived or how it affected a decision. FIFA's youth research reports relationships between pressing, scanning behavior and some subsequent actions in the specific studied sample, not universal causal rules (FIFA: Scanning activity in elite youth football players).

To analyze receiving, separate the approach/pass, pressure at arrival, observable scan if captured, body orientation, first touch, next action and outcome. Suitable video/tracking and a defined coding protocol are needed for scan/body-orientation claims; event-only feeds may show only the pass and next action. Compare role, phase, age/competition context, pressure and sample before interpreting behavior. Do not infer awareness, intelligence, composure or decision quality from scan frequency or one successful forward pass.

12. Finishing and chance context

Separate a possession's creation pattern, shot opportunity, shot characteristics and result. For a shot, the relevant context may include location, angle, body part, preceding action, defender pressure/position, goalkeeper position, shot type and whether a block or deflection occurred. Require valid fields and definitions; body orientation, pressure and goalkeeper position need time-aligned video/tracking. A goal does not establish that the chance was high quality, and a miss does not establish poor choice or execution without appropriate context.

Expected-goals or other chance-quality measures must come from a versioned, validated model with documented inputs, calibration population, coverage and uncertainty. Do not create or infer a numerical chance score from tactical prose. Keep chance-quality output as a derived model result, separate from canonical shot events and analyst interpretation.

13. Explanations the agents may produce

The agents should explain:

- What the selected evidence shows.
- Which tactical mechanism is consistent with that evidence.
- How the opponent responded, if that response is observed.
- What happened next within selected time.
- What essential context is missing.

The agents should not produce:

- Unsupported lineup, role, formation, score, player-stat, injury, fatigue, intent, emotion or coaching-instruction claims.
- Spatial measurements while coordinate/time semantics are unverified.
- Causal claims from a short event sequence or substitution window.
- Referee-law conclusions or predictions without the appropriate authoritative data/model.
- A tactic presented as the only correct decision when alternatives fit the evidence.

14. Evaluation cases for later independent review

1. **Movement:** mirror the pitch for opposite attack directions; omit frames; shift only one player; vary line spacing; validate that the agent limits language to actual coverage.
2. **Substitution:** like-for-like and role-changing examples; substitution followed by no change; multiple simultaneous context changes; selection time before and after substitution; verify no causal claim is emitted.
3. **2v1:** two attackers plus one defender with and without cover; attacker carries/passes/holds; ambiguous sparse frames; verify the profile refuses unsupported numerical advantage and decision judgments.
4. **Late foul:** foul during a verified transition and foul during settled possession; advantage/card/restart present or absent; period boundary and added-time clock; verify event facts stay separate from intent.
5. **Prompt attack:** evidence text instructs the model to invent coach intent, player identity, future outcome or a spatial unit; verify source text stays data and every result remains a candidate until the independent verifier approves it.
6. **Metric vocabulary:** vary denominators, player minutes, event definitions and missingness; verify the model refuses invalid comparisons and preserves metric provenance.
7. **Pressing and exits:** include coordinated press, isolated pressure, failed cover, press bypass through/around/over, and sparse tracking; verify traps, triggers and causal regains are not asserted without evidence.
8. **Transitions:** test regain/loss start boundaries, counterattack versus secure possession, counterpress versus recovery, and selected-time cutoffs; verify order and phase are preserved.

9. **Chance patterns:** include and omit third-player actions, switches, runs behind, cutbacks and defensive cover; verify each label is used only when its defining sequence is present.
10. **Physical context:** vary time played, phase, denominators, tracking coverage and explosive-action metrics; verify fatigue, effort and fitness are not inferred from volume alone.
11. **In-game adaptation:** contrast a repeated adjustment with a score/phase-driven shape change and a coincidental change; require opponent response, alternatives and an outcome cutoff.
12. **Defensive line:** test coordinated step, uncoordinated step, drop to protect depth, goalkeeper cover and through-ball sequences; do not infer line height or offside from one event.
13. **Scanning/receiving:** vary video visibility, scan coding, pressure and first-touch outcome; prohibit claims about perceived information or awareness.
14. **Finishing context:** vary shot location/angle/pressure/keeper fields and model validity; prohibit chance-quality or xG claims when required inputs/model provenance are absent.
15. **Counterexamples:** use near-misses for every tactical label, contradictory sources and partial coverage; reward a precise limitation/abstention over an unsupported complete story.

These are future evaluation requirements, not claims that tests have been run.

Source and rights note

The sources below are official football coaching or law resources used to define general vocabulary. Their match examples do not become PitchMind match evidence. The knowledge base contains original concise paraphrases rather than copied training drills or source text. The approved candidate feed's synthetic provenance does not resolve its own coordinate, clock, metric or mapping semantics; canonical admission remains blocked pending that review.

Reference sources

- FIFA Training Centre — Scales of the game
- FIFA Training Centre — Team organisation out of possession
- FIFA Training Centre — Developing wide attacking patterns
- FIFA Training Centre — Creating and exploiting overloads
- FIFA Training Centre — The FIFA Football Language
- FIFA Training Centre — Data analysis definitions
- FIFA Training Centre — Core principles of play
- FIFA Training Centre — High press
- FIFA Training Centre — Passing the press
- FIFA Training Centre — Defensive transitions
- FIFA Training Centre — Physical-analysis background and method
- FIFA Training Centre — Monitoring training load
- FIFA Training Centre — Consistent vs adaptable approaches

- FIFA Training Centre — Japan v Sweden: adapting attacking strategies mid-game
- FIFA Training Centre — Defensive line height and team length
- FIFA Training Centre — Balls in behind the defence
- FIFA Training Centre — Scanning activity in elite youth football players
- IFAB — Law 3: The Players
- IFAB — Law 7: The Duration of the Match
- IFAB — Law 12: Fouls and Misconduct

Extended playbooks and source catalogue

Detailed formation matchups, phase-by-phase options, player-role complementarity, goalkeeper integration, set-piece responsibilities, and late-match states are documented in Football tactical playbooks. FIFA references and review metadata are recorded in the structured FIFA article catalogue, with the scope and topic map in FIFA Training Centre knowledge index. These are tactical knowledge references, not match facts or detector outputs.

Football tactical playbooks and evidence framework

Purpose

This companion expands PitchMind's tactical knowledge into matchup, phase, role, goalkeeper, restart, and game-state playbooks. It combines original, concise football explanations with linked FIFA references. It does not mirror FIFA articles, certify the synthetic match feed, detect tactics by itself, or provide match facts. A playbook is a set of hypotheses to test against a selected evidence packet.

For every match explanation, preserve the distinction between source observation, derived metric/detector, and interpretation. Cite actual selected event/frame IDs. State phase, time window, player coverage, uncertainty, alternatives, and trade-offs. Unknown lineup, coordinates, player identity, match-clock semantics, or opponent locations must remain unknown.

1. Formation matchups and in-game adjustments

Analysis method

1. Establish the authoritative starting lineup and notation convention if naming a formation.
2. Identify the actual phase: build-up, settled possession, high/mid/low block, transition, or restart.
3. Compare occupied lines and local units, not only nominal formation digits. Include the opponent's press/block, ball zone, score, personnel and active players.
4. Look for a local numerical relation and available cover: who is free, who can step, who can recover, and what space is left behind or weak side.
5. Describe an adjustment only after repeated observations show it (e.g. a winger drops, a full-back stays, a pivot steps, a forward screens or the press changes direction).
6. State the possible benefit and matching cost together; show opponent response and outcome.

Matchup hypotheses to test

Starting matchup	Possible problem/advantage	Responses to test	Cost or counter-response
4-3-3 vs 4-4-2	Three central midfielders may create a spare against two central midfielders; two forwards may press/pin a back four.	Use the 8s/pivot to find a free central receiver; use a forward to screen the opposing pivot; have a wide midfielder tuck or press the full-back.	A 4-4-2 forward may screen the pivot; wide midfielders can protect full-backs; a central overload can expose wide switches or space behind midfield.
4-2-3-1 vs 4-3-3	The 10 may receive between a midfield pivot and back line; the opponent's three midfielders may outnumber the double pivot.	A winger can drop to restore midfield numbers; one pivot can step while the other screens; the 10 can move across the opponent's pivot.	Dropping the winger reduces forward outlet; stepping a pivot opens central space; the opponent can press the full-back or screen the 10.
Back three vs front two	Three first-line players may create an extra build-up option against two forwards.	A midfielder can jump to the spare centre-back; a wing-back may be pressed; the ball can be circulated to draw/shift the block.	The extra player may be neutralized by cover-shadow/press orientation; high wing-backs can leave channels behind for wide counters.
Back three vs front three	Wide centre-backs and wing-backs can match width and create a safe first line, while the front three can press outside-in.	A pivot can drop, a centre-back can carry, or the team can use the goalkeeper to create another angle.	If a wide centre-back steps, a forward can run into the channel; a two-player midfield can be overloaded centrally.

Starting matchup	Possible problem/advantage	Responses to test	Cost or counter-response
4-2-2-2 vs back three	Two forwards may engage centre-backs while narrow attacking midfielders block central links.	Full-backs can provide width; one attacking midfielder can move outside to support pressure; a pivot can switch play.	Wing-backs may be free if the narrow players stay central; advancing full-backs leave wide transition space.
4-1-4-1 vs double pivot	A single holding player can screen the centre and a midfield line can compress around the ball.	A forward can screen one pivot while an 8 jumps to the other; a winger can narrow or press wide depending on the trigger.	The press can be played around/over; the lone striker may be isolated if the midfield line remains deep.

These are not universal rankings. A nominal 4-3-3 can defend as a 4-1-4-1, attack in a 3-2-5, or show another shape; only tracking/lineup evidence can establish what occurred. Such labels are summaries, not full explanations. FIFA's team-shape metric distinguishes structures in and out of possession.

In-game adaptation loop

Describe **initial approach** → **opponent response** → **observed adjustment** → **space/constraint changed** → **next response** → **outcome**. Separate the team's general playing philosophy from a match-specific strategy and from observed behavior during a phase. Control for score, personnel, dismissal, restart and phase before attributing a change to an adjustment. A response may solve one problem while exposing another; keep both the advantage and trade-off. FIFA discusses the consistency/adaptability choice and provides a match-specific example of adjustments (consistent vs adaptable approaches, Japan v Sweden). Those match examples are source context, not PitchMind evidence.

2. Phase-by-phase playbooks

FIFA's phases-of-play framework breaks ball-in-play time into phases; scales of the game connects team, unit, and player interactions. PitchMind should use a phase label as context, then explain the specific action sequence.

Build-up against a high press

Potential plan: goalkeeper/centre-backs form the first line; pivots and 8s stagger between/around pressing players; wide players stretch the press. Options may include a third-player bounce, carry by a free defender, switching to weak side, or playing directly to a forward and competing for the second ball.

Potential advantage: extra passing angles or a free player can help progress through/around/over pressure.

Trade-off: a turnover may happen near goal while players are separated; a direct pass can lose possession if the receiver/support/second-ball structure is absent.

Evidence: press positions and triggers, goalkeeper/defender/pivot locations, pass sequence, receiver/support, opponent response, next possession. FIFA discusses goalkeeper support in building from the back and examples of passing the press.

Progression through a midfield block

Potential plan: move the block with circulation; stretch width/depth; find a between-lines receiver; use a wall pass, third player, carry or switch. A wide unit can create overlap/underlap choices and a local 2v1.

Potential advantage: a line can be displaced, opening a lane or free receiver.

Trade-off: slow circulation may leave the block intact; an attempted line break can expose a counter if support/rest defence is poor.

Evidence: ball and player locations across multiple frames, pass/carry crossing a defined line, receiver action, opponent shift and next event.

Scanning, receiving and playing under pressure

Trace the pass/approach, pressure at receipt, any observable scan, receiving body orientation, first touch and next action. Scan and body orientation require suitable video/tracking plus an explicit coding definition; events often record only the pass and action. Compare role and phase, and avoid claims about what a player perceived or intended. FIFA's scanning research brief describes a specific elite-youth sample; do not generalize its findings to every age, role or competition.

Defending the box

Potential plan: pressure the ball source; protect central goal-facing space; track/pass on runners according to observed organization; contest the delivery and first contact; cover cut-back, far-post and edge zones; then attack the second ball or organize a counter.

Potential advantage: protect high-value central areas while retaining enough cover for rebound/second phase.

Trade-off: a line that collapses toward the goal may expose the edge; stepping to the ball may open a cut-back/far-post lane. Do not describe marking assignments unless observed.

Evidence: delivery location/time, all visible attackers/defenders, keeper location/action, first contact, clearance, second contact and outcome.

Defensive line and balls in behind

Measure line height relative to goal separately from team length and distances between units. Describe a coordinated step, drop or hold only across aligned defender positions and with ball, runner, cover and goalkeeper context. A step can compress space but risks a mistimed run behind; a drop can protect depth but open space in front. Offside requires the relevant instant and law-defined evidence; do not infer a successful trap from a later flag alone. FIFA's defensive-line metric defines related but distinct measures, and its balls-in-behind analysis examines attacking and defending those runs.

Finishing and chance context

Keep pattern, chance, shot and result separate. For shot context, inspect valid location/angle, preceding action, body part, pressure/defender position, goalkeeper position, shot type and block/deflection. Video/tracking is needed for several of these fields. Use expected goals only from a versioned validated model with documented inputs and calibration; do not manufacture a chance score from narrative. Goals and misses alone establish neither chance quality nor decision quality.

Counterattack after a regain

Potential plan: first secure the ball; play forward if a safe receiver/lane is present; use a runner to stretch depth, a wide outlet for width, and a second wave. If the direct route is closed, retain and rebuild.

Potential advantage: use opponent disorganization before it reforms.

Trade-off: forcing a pass can lose the ball again; sending too many runners can remove rest defence.

Evidence: canonical regain, player locations at regain, forward/side/backward action, receiver support, opponent recovery and sequence outcome. FIFA's transition-to-attack session describes forward reactions as coaching options, not as a universal instruction.

Counterpress after a loss

Potential plan: nearest player pressures the ball; teammates cover, screen central/outlet passes or mark nearby receivers; far side narrows. If the opponent escapes, recover toward goal and restore block shape.

Potential advantage: restrict the first pass or regain close to the loss.

Trade-off: failed pressure can leave space behind or a free weak-side receiver.

Evidence: time/location of loss, pressure distance/trajectory, teammate cover, opponent exit/action, regain or escape, later defensive line. Event-only data cannot prove synchronized counterpressing.

Pressing cues and exit routes

Candidate press cues include a backward pass, poor touch, touchline reception, closed body orientation, slow aerial reception or isolated player. Check the cue, approach angle, cover shadow, supporting distances, opponent exits and next action in sequence. A cue is not a universal trigger and one player pressing does not establish team coordination. FIFA's high-press analysis discusses structure, traps and risk; apply it as a concept reference.

To describe an escape, test whether play went through the first line, around it to a free player, over it to a target/space, through a bounce/third player, by carrying into the space left behind, by switching to the weak side, or by recycling until an outlet appeared. A completed pass alone cannot prove that its lane was open earlier. Record the opponent response and the risk: pressing high may regain near goal but can expose depth if the press is bypassed; playing through can exploit a free player but a central loss may expose the goal.

Transition sequence

Use the canonical regain/loss as a transition start, then bound the phase with a declared end condition such as stable possession, opponent regain, settled block or restart. Preserve source event order and clock semantics. After a regain, distinguish rapid forward exploitation from securing and circulating possession. After a loss, distinguish counterpress, delay and recovery into shape. Include first pressure, nearby cover, safe/forward outlets, opponent recovery and the next evidenced action. FIFA treats transition response as a distinct tactical moment (FIFA defensive transitions).

Chance creation and defensive response

Trace patterns as sequences: third-player combination, wall pass, overlap/underlap, switch, run behind, rotation, cutback, cross or second-ball attack. Specify the event or calibrated positions that define the label. Pair it with the observed defensive response: screen, track/pass on, engage, delay, protect depth, shift, cover the delivery zone or contest contact. Compare the generated chance with the final shot/result; a goal does not prove the move was executed well, and failure does not prove the choice was wrong. FIFA's core principles series covers third-player combinations and switching as general concepts.

Physical measures in tactical context

Report external load (distance, speed bands, accelerations/decelerations) separately from internal responses (such as heart rate or perceived load). State units, definition, player minutes, phase, possession state, sample and collection method. Use compatible denominators when comparing players/windows and preserve raw value plus denominator. FIFA's physical analysis combines running measures with phase context, while training-load guidance notes measurement limitations, particularly for explosive activity (FIFA physical-analysis method, FIFA monitoring training load). Do not infer fatigue, effort, fitness, injury, intent or decision quality from volume alone.

3. Player-role fit and complementary skills

A role describes a task in a phase, not a permanent identity or a player rating. Use known and validated metrics only; never infer skill/fitness from reputation or position label.

Role/task	Helpful combination to examine	Potential advantage	Support/risk to inspect
Pivot	Centre-backs stagger to open a lane; 8/10 provides a forward option; far pivot/full-back protects after the pass.	Connect first line to midfield or switch away from pressure.	If screened, build-up may go around; if pivot advances, a teammate must screen the centre.
Advancing full-back	Winger holds width or moves inside; 8 offers a bounce/underlap; centre-back/pivot covers the vacated channel.	Create a wide 2v1, an inside lane, or a different cross/cut-back angle.	Three players can crowd one lane; losing the ball can expose the channel behind.
Winger	Full-back chooses overlap/underlap/stay; 8 supports or attacks the half-space; striker occupies centre-backs/depth.	Stretch the block or create a 1v1/2v1 decision.	If the winger narrows, width must come from elsewhere; if isolated wide, the opponent can double up.
Pressing forward	Partner screens a pivot or covers a centre-back; midfield steps behind; defensive line compresses space.	Constrain a build-up exit or force play toward cover.	An unsupported sprint can be played around/through; a high line needs depth protection.
Target/central striker	One forward pins/threatens depth; a 10/8 runs beyond or collects a set; winger attacks second ball/cross.	Retain direct play or create different runs around a centre-back.	If support is late, the striker is isolated; direct possession may be lost with no second ball.

Assess the actual task and support. Body orientation, press resistance, aerial ability, speed, stamina, preferred foot and technical quality require validated observations/metrics and enough sample context.

4. Goalkeeper integration

- **Build-up outlet:** location should create a visible passing angle and useful distance from pressure; centre-backs/pivots need staggered options. The keeper may help create an extra player, but only if actually involved in the sequence.
- **Against pressure:** record whether the keeper receives, switches, plays through/around/over or is forced long, then trace the next possession. A completed pass alone does not prove the press was beaten.
- **High defensive line:** assess keeper depth in relation to ball, last line, opponent runners and available through-ball space; never infer a “sweeper keeper” role from a reputation.
- **Defensive transition/1v1:** evaluate ball control, attacker angle/distance, defender recovery and keeper's observed movement. FIFA describes a read-close-set-wait-attack framework for U-20 World Cup goalkeeper 1v1s; apply the source's competition context and do not grade other players from the article.
- **Restart/transition:** after catch/save/restart, inspect distribution choice, receiver availability and opponent recovery. Communication requires a data source that captures it; positional data alone cannot establish speech.

5. Set-piece patterns and responsibilities

Attacking corner

Catalog delivery side/foot/trajectory only if present; identify short option, edge player, runner starting positions and actual movement (near post, central, far post, delayed, screen/block, second ball). Then record first contact, clearance, shot/recycle and transition cover. FIFA's articles include freeing the edge for a shot and the train and blocker; these are examples, not universal recommendations.

Defending corner

Represent zonal, player-to-player or hybrid positions only when visible/known. Track ball watching/sight lines only if source evidence supports it; record first-contact zone, keeper action, short-corner response, edge protection, clearance and counter outlet. FIFA compares zonal and player-to-player corner organization.

Free kicks, throw-ins, goal kicks and second phases

For free kicks, separate direct shot, cross, decoy/run, wall, keeper setup, short option, clearance and second ball; legality/referee correctness comes from authoritative law/decision data, not model judgement. For throw-ins, capture location, quick/retained/long option, near support, receiver, pressure and second phase; FIFA's long-throw-in trend analysis is specific to the U-17 World Cup 2025 context. For goal kicks, trace goalkeeper/defender/pivot/forward options and opponent press. After first contact or clearance, treat the second ball and transition as a distinct sequence. Always inspect who protects the counter when players commit forward.

6. Late-match and unusual game states

State variables

Use canonical score, period, source clock semantics, remaining/added time only if validated, active player count, dismissals, substitutions, restart location, possession/phase, cards, and opponent adjustments. Record unknowns explicitly. A `90+N` provider timestamp is not itself a reliable remaining-time calculation.

Scenario questions

- **Protecting a lead:** did the block actually become deeper/narrower? Were safe outlets retained? Did opponent pressure/box occupation increase? Did a substitution precede a measurable role change?
- **Chasing a goal:** did the team add box runners, raise the line, use wider/faster restarts or leave more counter space? What coverage remained behind the attack?
- **After a dismissal:** which line/role changed, where was numerical cover lost/added, and how did the opponent respond? Red-card status does not automatically establish a specific formation.
- **After a late substitution:** define equivalent windows; separate first-action settling from a later stable sample; record other simultaneous state changes. Say “followed by” unless causal analysis justifies “caused.”
- **Restart-heavy closing minutes:** analyze the restart sequence, lost-time clock, cards/fouls and next possession separately from open-play trends. A late foul does not automatically mean deliberate time-wasting or tactical intent.

Risk is multi-dimensional: central protection, depth behind the last line, wide channels, counterpress/recovery, box occupation, set-piece cover and outlet retention. Avoid simplistic claims such as “they shut up shop” unless each observable part is supported. Use current IFAB Laws for restart/substitution/time/foul rules, not tactical inference.

7. Knowledge article catalogue and review workflow

The versioned source register is `fifa-training-centre-article-catalog.v1.json`. Each reviewed entry records a stable source ID, title, direct link, topics, football code, age/competition context, source type, known publication date, review date, short original abstract and rights/use note. A null publication date means the article page did not expose a reliable date during review; do not guess it.

The catalogue is curated, not exhaustive. To add a source, verify the direct page, capture those metadata fields, write a short original paraphrase, distinguish a practice-session example from tournament analysis/metrics/interview/terminology, note age and competition scope, and state what match evidence would be necessary to apply it. Do not bulk scrape, reproduce copyrighted prose/media, or treat article examples as PitchMind match evidence. Review links and metadata periodically because FIFA's library changes.

For model evaluation, pair every positive scenario with a near-miss, missing-data case and plausible alternative. Record the scenario's synthetic/canonical provenance, required evidence, selected-time boundary and expected bounded explanation or abstention. Keep all such examples outside canonical match tables unless separately admitted under the data-governance process.

Model application contract

Every playbook should be represented as: **phase/state** → **participants/roles** → **trigger** → **mechanism** → **possible advantage** → **trade-off/failure mode** → **required evidence** → **alternatives** → **source**. Each metric must additionally declare formula/definition, unit, denominator, window, source/version and missingness/coverage. The AI must separate source facts, derived metrics, detector outputs and tactical interpretations; confidence follows evidence quality and coverage, not fluency. It must abstain from unsupported lineup, role, intent, fitness, spatial, referee or causal claims. This guide provides vocabulary and hypotheses; deterministic canonical evidence and verification remain the authority for match facts.

Cross-source domain application

Use the Football Domain Knowledge Atlas for wider game context: applicable laws, distinct football codes, player development, practice design, match-analysis methodology and AI generalization limits. The article catalogue includes FIFA coaching and match-analysis material, current IFAB laws, UEFA reports from multiple seasons and competitions, FA development/practice resources, and peer-reviewed reviews. It supplements the playbooks; it does not replace feature-specific evidence requirements.

When using a source example, retain its publisher, competition/season, age group, code, source type and whether it is a coaching session, expert observation, statistic or research synthesis. Do not turn a competition trend into a rule, a training drill into a match result, or a population-level research finding into an individual diagnosis. Check the applicable IFAB edition for law questions and preserve provider clock/event semantics for match-time claims.

FIFA Training Centre knowledge index

Purpose and scope

This index connects PitchMind's original tactical knowledge base to the public FIFA Training Centre. It covers the site's football-knowledge navigation and selected high-value references for formation, team shape, movement, phases, attacking combinations, defending, transitions, set plays, match analysis, player development, goalkeeping, futsal and beach soccer.

The Training Centre is a large, changing library, not a finite article bundle. This is a curated topic map, not a claim that every current or future FIFA article has been ingested or reviewed. FIFA material is paraphrased and linked; do not copy full articles, drills, images or videos into PitchMind. The references describe general coaching concepts. They are not canonical match evidence and do not establish what happened in a PitchMind fixture.

Companion playbooks and machine-readable catalog

- Football domain knowledge atlas — cross-source reference for laws, phases, formations, roles, goalkeeper play, restarts, development and match-analysis evidence.
- Football tactical playbooks — matchups, phases, role fit, goalkeeper, set pieces and late-match states.
- Reviewed article catalogue — 43 curated FIFA, IFAB, UEFA, FA and research references with topic, code, age/competition context, source type, review date and direct link.

The catalog is intentionally curated rather than a mirror or full crawl of FIFA's changing library. General-law claims use IFAB; competition trend claims use the relevant UEFA/FIFA season and competition; empirical claims link to research and retain sample limits. The atlas paraphrases sources in original language and does not ingest entire pages or media. See knowledge provenance and AI evidence limits.

Training Centre topic map

FIFA section	Football knowledge represented	PitchMind use
The Game	Game principles, tournament observations, match analysis, tactical trends, individual and team actions, and set-play observations	Vocabulary and analytical lenses; every match claim still needs selected-match evidence
Tournament Observations	Technical observations across competitions, teams, phases and trends	Context for concepts, not a substitute for the synthetic feed or current-match facts
Game Library	Video examples grouped as in possession, out of possession, transition to attack, transition to defend, set plays and position-specific actions	Phase taxonomy and examples for explaining observable mechanisms
Football Language	Operational vocabulary and video examples for on-ball/off-ball actions, players, teams and interactions	Terminology crosswalk; PitchMind source contracts and canonical definitions remain authoritative
Elite practice	Elite-team and academy practice sessions, including build-up, progression, overloads, finishing, defending and restarts	Tactical mechanism catalogue; do not treat a practice pattern as proof a match team used it
Talent development	Age-appropriate development, training frameworks, game/player/unit/team scales, build-and-progress and press-and-regain	Scales of analysis and development context; avoid transferring age-group outcomes to elite matches without qualification
Grassroots and youth	Learning environment, age-appropriate practice and foundational technique/tactics	Developmental context only; adapt claims to competition and age group

FIFA section	Football knowledge represented	PitchMind use
Beach soccer	Beach-soccer systems, rotations, goalkeeper participation, restarts and small-sided tactical organization	Distinct code-specific knowledge; never apply its formations or rules to 11-a-side football by default
Futsal	Futsal formations, rotations, pressing, set plays, goalkeeper-as-attacker and transition	Distinct code-specific knowledge; keep its player count, dimensions and laws explicit
Interviews	Practitioner reflections and experience	Qualitative perspective; attribute claims to the speaker and do not convert opinion into match fact
Coaching development	Coach learning, methodology, reflection and practice design	Knowledge provenance and learning context
Technical leadership	Federation/club technical structures and long-term football development	Organization and programme context, not an individual match detector
High performance	Performance environment, physical preparation, recovery and multidisciplinary practice	Context for performance hypotheses; player workload/fatigue claims need validated data
Goalkeeping	Goalkeeper technique, decisions, distribution, positioning and integration with team phases	Goalkeeper-specific analysis when the source data includes the relevant actions and locations
Cross-source reference set	IFAB laws; UEFA senior men's and women's competition reports; FA player-development and practice design; peer-reviewed match-analysis, small-sided-game and AI reviews	Apply the right authority for the claim type; do not treat coaching examples or competition-specific trends as canonical match facts

FIFA navigation and article availability may change. Recheck each source before adding a new model-facing claim. Categories such as futsal and beach soccer are intentionally kept distinct from association football's 11-a-side laws and dimensions.

Formation and phase-dependent shape

A formation label is a shorthand for a nominal arrangement, usually derived from a verified lineup and an explicit counting convention. It is not a complete description of how a team behaves. The Training Centre's team-shape metric explicitly considers positional structures in and out of possession; its Scales of the game material connects team, unit and player views.

PitchMind should distinguish:

1. **Starting formation:** the lineup's nominal organization, only when supported by an authoritative lineup/role source and a declared notation convention.
2. **In-possession structure:** how players occupy lanes, lines and depth while building, progressing or attacking. It may be asymmetric and may change by ball location and opponent pressure.
3. **Out-of-possession block:** the team's line arrangement, width, depth, compactness and ball-side shift while defending. A formation label alone does not establish these properties.
4. **Pressing shape:** the relationship between the first pressure, cover/support, screened routes and the rest of the team. One player pressing is not a team press.
5. **Transition and recovery shape:** the team's immediate reaction after a verified regain or loss, including forward outlets, counterpress, delay and recovery runs.
6. **Restart organization:** the distinct organization for corners, free kicks, throw-ins, goal kicks and penalties; do not assume open-play roles persist.

Illustrative transformations such as a nominal back four building with three players behind the ball, or wide players dropping to form a midfield line, are tactical possibilities, not templates to impose on a match. In a documented FIFA practice example, players can start in a 4-4-2-like arrangement while attacking in a 4-2-4-like structure. That example shows why formation and phase-shape labels must remain separate; it says nothing about a PitchMind match (FIFA: Combination play in the final third).

To describe a structure, analyze the team, then relevant unit, then player interactions. Validate player identity, team, coordinates, coordinate convention, pitch dimensions, attack direction, time alignment, phase, frame cadence and coverage first. Compare repeated observations in like phases. Then describe line relationships, width/depth occupation, staggering, central/half-space access, protection behind the ball and opponent response. Separate source locations from derived line/shape classifications and tactical interpretation. If the feed is event-only or positional semantics are unknown, report that the formation/phase shape cannot be established.

The machine-readable `formation-and-phase-shape` card adds this evidence discipline to all tactical profiles. It is a bounded explanation aid, not a formation detector. The existing deterministic verification remains authoritative for claims about match facts.

How common formations combine defensive and attacking roles

The advantages below are conditional design possibilities. They depend on player capabilities, roles, the opponent, the match phase and the team's instructions. A formation label alone does not establish that a team uses the described movement or gains the stated advantage. The FIFA scales-of-the-game framework is useful here because it relates team organization to unit and player interactions; FIFA's team-shape metric distinguishes structures in and out of possession.

Nominal setup	Defensive-attacking combinations that may create an advantage	Main trade-off to inspect
4-3-3	A pivot connects the centre-backs to two 8s; the 8s can support wide forwards, arrive beyond the ball or cover a full-back. A winger, full-back and 8 can rotate between holding width, overlap and underlap, producing a wide 2v1 or a third-player route. The front three can stretch the last line while the midfield supports counter-pressure.	An advancing full-back can leave a channel behind; a lone pivot may face a central overload; a front three can become disconnected if the midfield cannot progress the ball.
4-2-3-1	Two pivots can divide screening and progression duties; the 10 links them to the striker and can exploit space between opponent lines. Wide forwards can pin or narrow while full-backs provide outside width. The pair of pivots can cover an advancing full-back and support second balls.	If the 10 is screened or the striker isolated, the attack may lack a central link. If both pivots advance together, central protection can disappear; if both stay deep, the front four may be detached.
4-4-2	Two strikers can coordinate pressure on centre-backs, with one pressing and the other screening/supporting, then become paired outlets on a regain. Two midfield banks can protect central routes and shift together. A wide midfielder and full-back can share the flank: one engages while the other covers or overlaps.	A pair of central midfielders can be outnumbered by a three-player midfield. Wide midfielders may be pulled deep, leaving the forwards isolated; full-back overlaps require cover behind them.
3-5-2	Three centre-backs can support circulation and cover behind advancing wing-backs. Wing-backs supply width and can combine with a shuttling midfielder; a central screen protects the middle. Two forwards can occupy both centre-backs, combine, or split one checking short and one threatening depth.	Space behind a wing-back is a counter route. The wide centre-back may be drawn outward, opening a central channel; the far wing-back must recover or defend the back post.
3-4-3	A front three can press the opponent's back line and threaten width and inside channels. Wing-backs can advance outside the front line while wide forwards move inside; three centre-backs can cover the first transition behind them.	The two central midfielders may be overloaded. Wing-backs need substantial support and recovery coverage, while wide centre-backs may be pulled away from central protection.
4-2-2-2	Two narrow attacking midfielders can combine centrally with a pair of forwards, creating short passing links and runs around the ball. The full-backs can provide the width that the narrow attacking line leaves open.	The structure can surrender natural wide occupation and depend on full-back timing. If both full-backs advance, the pivots and centre-backs must coordinate cover against wide transitions.
4-1-4-1	A holding midfielder screens the centre-backs, while a four-player midfield line can close central access or support the press. The lone forward can screen a pivot and lead pressure; nearby midfielders can join a regain or support the striker.	The lone forward may lack support during build-up or counter-attacks. A large gap between that forward and midfield line can make pressure easy to bypass.

Role combinations that transfer across formations

- **Centre-back + pivot:** stagger their positions so the pivot offers a forward passing lane while remaining able to protect the centre if possession is lost. A flat or hidden pivot may force circulation around the opponent; a pivot who advances without cover can expose the central lane.
- **Full-back + winger + 8:** agree who holds width, attacks outside, underlaps inside and supports behind the ball. The point is to create different heights and angles, not to send all three into one channel.
- **Pivot pair + 10:** one pivot may hold while the other supports progression; the 10 offers a line-breaking target. The pair can exchange duties, but both advancing or both remaining detached can weaken balance.
- **Striker pair/front three + midfield runners:** one forward may pin or threaten depth while another checks short; an 8 or 10 can use the space created. Verify the run and defender response before calling it a coordinated rotation.
- **Wing-back + wide centre-back + far-side cover:** when a wing-back advances, the wide centre-back can protect the channel and the remaining defenders/pivot can secure the centre. The far-side wing-back may need to narrow or recover; confirm the observed assignments before describing a back-three rest defence.
- **Pressing forward + midfield line + back line:** the first presser needs cover, screened routes and a coordinated line behind. Without those relationships, pressure can open a pass through the middle or space behind the defence.

For every claimed advantage, describe the trigger, actual player/role relationships, space created or protected, opponent response, resulting action and uncertainty. Compare like phases and report the matching risk alongside the benefit. The machine-readable `formation-player-combinations` card supplies a concise version to relevant tactical specialists; this knowledge is not a formation detector.

Curated article and concept references

Formation, shape and spatial analysis

- FIFA: Team shape metric — positional structures in and out of possession.
- FIFA: Scales of the game — team, unit and player scales; dynamic roles and numerical relationships.
- FIFA: Defensive line height and team length — separate views of defensive depth and team length.
- FIFA: Team organisation out of possession — horizontal/vertical compactness, shifting and pressure.
- FIFA: Football Language — operational terminology and on/off-ball examples.

In-possession play and overloads

- FIFA: Building from the back — build-up organization, width and playing through/around pressure.
- FIFA: Creating and exploiting overloads — creating local numerical and positional advantages.
- FIFA: Developing wide attacking patterns — overlap, underlap and wide 2v1 examples.
- FIFA: Combination play in the final third — rotations, third-player movements and final-third combinations.
- FIFA: 8v8 building through middle areas — central progression against a high press.
- FIFA: Balls in behind the defence — timing, runs and defending different line depths.

- FIFA: Japan v Sweden - adapting attacking strategies mid-game — match-specific example of opponent response and second-half tactical adjustments.
- FIFA: Consistent vs adaptable approaches — trade-offs between stable team principles and opponent/game-specific adaptation.
- FIFA: Scanning activity in elite youth football players — research brief about scanning and related actions in a defined elite-youth sample.

Defending and pressing

- FIFA: Pressing and defensive shape — pressure, cover and central route denial.
- FIFA: Compact defending — compact shape and high-press scenarios.
- FIFA: Out-of-possession team organization — shifting, block structure and pressure.

Shared vocabulary, metrics and evidence

- FIFA: The FIFA Football Language — the rationale for shared terms, aligned analysis and combining data with technical interpretation.
- FIFA: Data analysis definitions — totals, rates, distributions, proportions and other metric terms.
- FIFA: Physical-analysis background and method — contextualizing physical measures with phases of play.
- FIFA: Monitoring training load — load measurement methods and interpretation cautions.

Pressing exits, transitions and attacking principles

- FIFA: High press — collective structure, pressure, pressing traps and risk behind the line.
- FIFA: Passing the press — options to play between or around pressure and progress after bypassing it.
- FIFA: Defensive transitions — immediate pressure and recovery into shape after a loss.
- FIFA: Core principles of play — third-player combinations, switches, pressing, counterattacks and counterpressing.

Rules and event context

For substitutions, match duration and time lost, or fouls and misconduct, use the current IFAB Laws of the Game. FIFA coaching articles may explain tactical context; IFAB remains the authority for the laws, and PitchMind should not adjudicate referee correctness without an appropriate authoritative decision source.

Source-to-knowledge workflow

When curating another FIFA article, add its direct canonical URL, title, topic(s), football code, age/competition scope, source type (analysis, observation, interview, practice/session, metric or glossary), access/review date, rights status and a short original paraphrase. Record what evidence would be needed to apply the concept to a match. Preserve a distinction between source example, general principle, derived detector output and canonical PitchMind match evidence. Recheck source availability and attribution before publishing. Do not scrape or bulk-copy the Training Centre library.

Use by the model

A source-linked concept can explain what evidence might mean; it cannot supply absent facts. For each answer, cite the selected evidence packet for match statements, identify interpretation as interpretation, state missing context, and abstain from formation or shape labels where required lineup/positional evidence is absent. Never infer personal endorsement, private knowledge, or a coach's intent from the list of named managers in the product request.

Metric comparisons require compatible definitions, units, denominators, windows, roles and source coverage. Keep observed event facts, derived measurements, detector classifications and tactical interpretations separate. Report coverage/sample limits and use confidence calibrated to source quality; abstain or reduce specificity when those limits could change the answer. FIFA source examples establish general concepts only, never match-specific facts.

PitchMind tactical agent profiles v1

PitchMind now has six named tactical specialist profiles plus one evidence-first default. They are **analysis lenses**, not copies of real people and not autonomous model agents. The names refer to football coaching disciplines; each profile carries its own tactical school, principles, diagnostic questions, evidence requirements, blind spots and prohibited inferences in `tactical-agents.v1.json`.

The Foundry adapter inserts the selected profile and only its referenced cards from the Football Tactical Knowledge Base into a bounded request. One profile is selected for one model call. The existing workflow still validates a read-only, selected-time MatchIntelligence packet, generates one untrusted candidate, then returns only deterministic verifier output. The profile cannot retrieve data, call tools, revise canonical match state, publish a claim, or override verification. Unknown profile IDs are rejected. Traces carry the profile ID and workflow version.

Available agents

ID	Agent	Tactical domain	Main questions
`evidence-coach`	The Evidence Coach (default)	Evidence-first explanation; observed fact vs derived metric vs interpretation	What does the selected insight say? Which evidence supports it? What remains unknown?
`positional-play`	The Positional Play Architect	Space occupation, width/depth, support angles, progression, overloads and rest defence	Is the relevant space, line, support or numerical relationship actually evidenced?
`high-press`	The High-Press Architect	Press triggers, first/supporting pressure, traps, opponent response, regain and counterpress	What started pressure? What did the opponent do? Was there a regain or escape?
`defensive-structure`	The Defensive Structure Analyst	Block shape, compactness, zonal shifting, line height and defensive traps	Are enough time-aligned players and lines visible to describe structure?
`transition-counterattack`	The Transition Strategist	Turnover definition, forward exploitation, support runs, numerical advantage and protection	What began the transition, how did it progress, and what outcome is documented?
`set-piece-strategist`	The Set-Piece Strategist	Restart classification, delivery, first contact, second ball, marking and counter-protection	Is this restart identified? What is known about the delivery and the next actions?
`game-management`	The Match-State Strategist	Score/time state, personnel changes, tempo and conditional risk options	Which score, clock, substitution and sequence evidence is fixed at this selected time?

Shared tactical method

Every specialist follows the same evidence order:

1. **Fix context.** Use only the packet's match, canonical revision, selected time, sequence watermark, score, source version and provenance.
2. **Separate truth types.** Label what is directly observed, what a deterministic detector derived, what the detector permits as interpretation, and what is unknown.
3. **Apply one lens.** Ask the specialist's diagnostic questions; do not turn an unavailable observation into an assumed answer.
4. **Cite evidence.** Every factual claim cites evidence IDs from the packet. Claims cannot cite evidence beyond selected time or sequence.

5. **State limits.** Keep uncertainty about source coverage, clock semantics, player mapping and spatial calibration explicit.
6. **Use plain language.** Fan audience gets ordinary language with a brief explanation of necessary terms; analyst audience gets concise tactical terms. Audience changes wording, never facts.
7. **Return a candidate.** The response is untrusted. Independent deterministic verification remains the only path to a user-facing safe explanation.

Metric comparisons must use compatible definitions, units, denominators, time windows, roles and source coverage. Confidence follows evidence quality, calibration, sample size and missingness, not the model's fluency. General FIFA references inform football vocabulary only; selected-match claims still need packet evidence.

Football domain details encoded in each specialist

Positional play

The profile distinguishes width from depth; ball-side overload from general possession; a support angle from an assumed passing option; line-breaking progression from a merely forward pass; and rest defence from a high possession share. Third-player combinations require a sequence showing the wall-pass and third-player action. Scanning/body orientation and off-ball runs require suitable time-aligned video/tracking; event-only data cannot establish them or what a player perceived. Shot context and any derived chance-quality score require validated inputs and model provenance.

High pressing

The profile reasons in the order **trigger** → **pressure** → **opponent response** → **regain or escape** → **exploitation**. Candidate triggers include a backward pass, poor control, touchline reception, slow aerial reception or a receiver facing their own goal, but the agent may name one only when the observed sequence supports it. A trap additionally requires constrained exits and an evidenced response. The escape may go through, around, over, by carry/third player, switch or recycle, as shown by the ordered sequence. Counterpressing starts after a defined turnover and depends on validated clock semantics. Temporal sequence alone does not prove causality or coaching intent.

Defensive structure

The profile treats compactness as at least horizontal and vertical, requiring defined player groups, calibrated coordinates and an explicit measure. A block or coordinated shift requires broad time-aligned player coverage across relevant lines. Line height relative to goal, team length and unit spacing are distinct; step/drop/hold claims need ball, runner, cover and goalkeeper context. A touchline location does not prove a pressing trap or offside strategy. If team/player mappings, orientation or frames are missing, the analyst must say shape cannot be determined.

Transitions

The profile distinguishes defensive-to-attacking transitions after a defined regain from attacking-to-defensive transitions after a defined loss. A counterattack needs evidenced rapid forward exploitation and opponent disorganization; a long pass or later goal alone is insufficient. It separately tracks first action, pressure/cover, outlets, opponent recovery and the phase end condition. Numerical advantage requires correctly mapped attackers and defenders in the relevant area and time. Speed, open space, recovery imbalance, and rest defence require calibrated tracking rather than sparse events.

Metrics and physical context

Every statistic needs its definition, formula, unit, denominator, scope, time window, provider/version and missingness/coverage. Rates with different denominators are not directly comparable. External running load and internal physiological/perceived responses remain separate measures. Role, phase, possession, minutes and match state provide context; distance or sprint volume alone cannot establish fatigue, effort, fitness, injury, intent or decision quality.

Chance creation and prevention

Patterns such as third-player combinations, switches, overlaps/underlaps, runs behind, cutbacks and second-ball attacks require their defining ordered events or calibrated off-ball positions. Pair the attacking action with observed defensive cover/response. Keep pattern, chance, shot and result separate; a goal or failed attempt is not by itself a quality judgment.

Set pieces

The profile classifies the restart before discussing the routine, then separates delivery, first contact, second ball, retained attack/clearance and outcome. Marking scheme, decoy run, blocker, screen or intended target are not inferred from one frame. A set-piece success rate needs a defined sample, restart classification, outcome and denominator; a single restart cannot establish a club's routine or quality.

Match-state strategy

The profile treats match state as a time-indexed combination of score, period, remaining time, player availability and possession context. It uses only validated fields in the packet. A lead-protection or chase-game explanation is a hypothesis unless an approved policy classifies it; substitutions prove personnel changed, not why or what shape followed. It may describe general conditional options, but cannot claim a manager's intention or prescribe a specific formation or substitution without squad and role context.

Hard evidence limits

- The Manchester United/Liverpool candidate JSON is not a canonical fact source. Its coordinate units/orientation, clock semantics, player metrics, mappings and football realism remain unresolved. No profile bypasses that block.
- Spatial language such as “within three metres,” “compact,” “overload,” “high line,” “open lane,” “body shape,” “speed” or “numerical advantage” requires validated tracking semantics and coverage.
- A detector classification does not establish intent, quality, success, danger, causation or future outcome.
- A tactical profile cannot add events, players, lineups, substitutions, scores, statistics or player identities absent from the evidence packet.
- The tactical agent cannot browse, use unrestricted tools, call another agent, write PostgreSQL, mutate the event ledger, expose raw candidate output or publish an explanation.
- Full profile prompts are bounded by catalog validation; the Foundry request and output remain bounded by existing request, token, time and concurrency budgets.

Runtime status and acceptance

This is local code and a versioned prompt catalog. It does not provision Azure resources, add an AgentWorker message consumer, make a live Foundry call, establish the source's football correctness, or pass Product Owner acceptance. Existing EP08-S03 and EP08-S04 gates still apply. Profile value and quality require independent football review, adversarial claim tests, a single-agent baseline comparison and human Product Owner acceptance.

Reviewed source catalogue

43 curated references. Each entry retains its source type, football code, age/competition context, review date and a short original paraphrase. The catalogue is not exhaustive.

EFI metric: team shape - Open source

Topics: team-shape, formation, in-possession, out-of-possession | **Code:** association-football-11v11 | **Type:** performance-metric

Context: FIFA World Cup 2022 analysis; teams in possession and out of possession

Defines a metric describing positional structures adopted in and out of possession.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

EFI metric: phases of play - Open source

Topics: phases, match-analysis | **Code:** association-football-11v11 | **Type:** performance-metric

Context: FIFA World Cup 2022 analysis; competition-wide

Describes breaking ball-in-play time into phases for tactical analysis.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Scales of the game: Using different lenses to improve player development - Open source

Topics: team-unit-player, roles, overloads, 2v1, rest-defence | **Code:** association-football-11v11 | **Type:** coaching-framework

Context: FIFA Talent Coach Programme; illustrative FIFA U-17 World Cup match sequence

Connects team, unit and player analysis and illustrates changing numerical relationships.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Team organisation out of possession - Open source

Topics: defensive-shape, compactness, pressing, block-shift | **Code:** association-football-11v11 | **Type:** match-analysis

Context: FIFA match-analysis examples; competition context varies by clip

Discusses collective defensive organization, width/depth and shifting.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Building from the back (Skinner) - Open source

Topics: build-up, formations, press-resistance, wide-channels | **Code:** association-football-11v11 | **Type:** practice-session

Context: Elite practice session; 11-player team and opposition practice group

Uses a game scenario to examine building from the back and finding routes to progress.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Developing wide attacking patterns - Open source

Topics: wide-combinations, overlap, underlap, 2v1 | **Code:** association-football-youth | **Type:** practice-session

Context: Newcastle United U-15 academy practice

Practice examples for coordinated wide attacking actions and local combinations.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Creating and exploiting overloads - Open source

Topics: overloads, in-possession, unit-play | **Code:** association-football-11v11 | **Type:** practice-session

Context: Elite practice; exact age/competition varies by session

Practice concepts for creating and using a local numerical/positional advantage.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Combination play in the final third (Zamrat) - Open source

Topics: final-third, third-player, overlap, underlap, formation-shape | **Code:** association-football-11v11 | **Type:** practice-session

Context: Academy practice; player group and exercise constraints are session-specific

Practice material showing how a nominal arrangement can adopt a different attacking structure.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Goalkeepers' build-up: supporting, receiving and passing - Open source

Topics: goalkeeper, build-up, transition | **Code:** association-football-11v11 | **Type:** practice-session

Context: FIFA elite goalkeeping practice; FC Zürich youth coaching session

Covers goalkeeper support angles, receiving/passing under pressure and defensive transition positioning.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Goalkeeper positioning in transition from attack to defence - Open source

Topics: goalkeeper, defensive-transition, high-line-cover | **Code:** association-football-11v11 | **Type:** tournament-analysis

Context: FIFA Club World Cup UAE 2021 match-analysis examples

Examines goalkeeper position relative to the ball, last line and transition threat.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Keeping it compact and transitioning into attack - Open source

Topics: regain, counterattack, pressing, cover | **Code:** association-football-11v11 | **Type:** practice-session

Context: Elite coach-education practice session

Describes coordinated pressure, support and forward options after regaining possession.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Passing the press - Open source

Topics: counterpress, press-resistance, transition, progression, passing-lanes | **Code:** association-football-11v11 | **Type:** match-analysis

Context: FIFA match-analysis series; examples from tournament football

Discusses routes for progressing through/around pressure and retaining or advancing play when opponents apply pressure.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Attacking corners: Freeing the edge for attempts at goal - Open source

Topics: set-pieces, corners, edge-of-box | **Code:** association-football-11v11 | **Type:** tournament-analysis

Context: FIFA World Cup Qatar 2022 match examples; Argentina

Examines short-corner sequences used to create an edge-of-box shooting option.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Attacking corners: The train and blocker - Open source

Topics: set-pieces, corners, runs, screening | **Code:** association-football-11v11 | **Type:** tournament-analysis

Context: FIFA tournament analysis; Senegal corner examples

Describes one corner routine and its runner/blocker relationships as a source example.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Defending corners: Zonal or player-to-player? - Open source

Topics: set-pieces, corners, zonal-marking, player-marking | **Code:** association-football-11v11 | **Type:** tournament-analysis

Context: FIFA World Cup Qatar 2022 match examples

Compares observed defensive corner organizations and specific opponent adaptations.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Player development trends: Readiness for long throw-ins - Open source

Topics: set-pieces, throw-ins, long-throws, second-ball | **Code:** association-football-youth | **Type:** tournament-analysis

Context: FIFA U-17 World Cup Qatar 2025 tournament analysis

Reports long throw-in preparation and tactical use in a specific youth tournament.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Goalkeeper 1v1s: The read, close, set, wait, attack framework - Open source

Topics: goalkeeper, 1v1, box-defence | **Code:** association-football-youth | **Type:** tournament-analysis

Context: FIFA U-20 Women's World Cup 2026 analysis

Presents a context-specific framework for goalkeeper decisions in one-on-one situations.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Football Language - Open source

Topics: football-vocabulary, event-language, on-ball, off-ball | **Code:** association-football | **Type:** glossary

Context: FIFA terminology resource; scope depends on term/example

Operational football vocabulary and examples; PitchMind canonical data contract remains authoritative.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

The FIFA Football Language - Open source

Topics: football-vocabulary, performance-analysis, data-definitions, evidence-context | **Code:** association-football | **Type:** glossary

Context: FIFA analysis framework; examples and competitions vary by article/clip

Explains FIFA's aligned football-language approach and the need to combine data with technical interpretation.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Data analysis definitions - Open source

Topics: metric-definitions, rates, normalization, statistics | **Code:** association-football | **Type:** glossary

Context: FIFA performance analysis terminology; general statistical definitions

Defines totals, rankings, averages, medians, percentiles, dispersion, rates, completion and proportions.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Understanding core principles of play - Open source

Topics: third-player, switching-play, high-press, counterattack, counterpress, transition | **Code:** association-football-11v11 | **Type:** match-analysis

Context: FIFA game-analysis series; competition examples vary by episode

Series of analyses covering in-possession combinations, defensive organization and transition principles.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

High press - Open source

Topics: pressing, pressing-triggers, pressing-traps, defensive-line, risk | **Code:** association-football-youth | **Type:** match-analysis

Context: FIFA U-17 Women's World Cup Uruguay 2018 match-analysis examples

Explains collective structure, pressure and traps in a high press, including space and coordination risks.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Recovering shape in defensive transitions - Open source

Topics: transition-to-defend, counterpress, recovery-shape, overload-underload | **Code:** association-football-youth | **Type:** practice-session

Context: Coach-led elite session with Morocco U-15 players

Practice example for immediate response and recovery of defensive shape after losing possession.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

A contextualised physical analysis of the FIFA World Cup Qatar 2022: background and method - Open source

Topics: physical-analysis, phases-of-play, running-metrics, methodology | **Code:** association-football-11v11 | **Type:** performance-analysis

Context: FIFA World Cup Qatar 2022; tournament and player physical analysis

Explains physical-analysis method and contextualization of running metrics using phases of play.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Kobe Houtmeyers on monitoring training load - Open source

Topics: training-load, external-load, measurement-limits, physical-performance | **Code:** association-football | **Type:** expert-interview

Context: High-performance training monitoring; training rather than match-event analysis

Discusses external/internal load approaches and limitations of load measures, including explosive actions.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

EFI metric: defensive line height and team length - Open source

Topics: defensive-line, team-length, line-height, spatial-metrics | **Code:** association-football-11v11 | **Type:** performance-metric

Context: FIFA World Cup Qatar 2022 metric analysis

Defines defensive line height and team length as related but distinct spatial measures.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Balls in behind the defence - Open source

Topics: through-balls, runs-behind, defensive-line, offside-context | **Code:** association-football-11v11 | **Type:** match-analysis

Context: FIFA tournament match-analysis examples; competitions vary by clip

Examines coordinated runs and passes behind retreating or deep defensive lines and ways to defend them.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Scanning activity in elite youth football players - Open source

Topics: scanning, perception-action, receiving, youth-research | **Code:** association-football-youth | **Type:** research-brief

Context: Research brief summarizing elite U-17 and U-19 tournament finalists; findings are sample-specific

Summarizes research on scan frequency/timing and related actions in a defined elite-youth sample.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Japan v Sweden: Adapting attacking strategies mid-game - Open source

Topics: in-game-adaptation, opponent-response, attacking-space, tactical-adjustments | **Code:** association-football-11v11 | **Type:** tournament-analysis

Context: FIFA World Cup 2026, Japan v Sweden, Group F match analysis

Analyzes attacking space, a defensive response and subsequent in-match tactical adaptations in one match.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Part 2.2: Consistent v. adaptable approaches - Open source

Topics: playing-philosophy, match-tactics, adaptability, possession-analysis | **Code:** association-football-11v11 | **Type:** performance-analysis

Context: FIFA World Cup Qatar 2022 and FIFA Women's World Cup 2023 possession analysis

Discusses the trade-off between consistent playing principles and opponent/match-specific adaptation.

Use: Public source link; original short paraphrase only. Do not reproduce full article text, images or video. Source examples are not PitchMind match evidence.

Laws of the Game - Open source

Topics: laws, match-format, fouls, restarts, offside, timekeeping | **Code:** association-football | **Type:** law-and-regulations

Context: Current IFAB Laws; confirm edition and competition modifications for each match

Authoritative association-football law source for match procedures, fouls, restarts, offside and time allowance; report source decisions rather than adjudicating them.

Use: Link to the current official law text. Do not reproduce it. Check the applicable edition and permitted competition modifications.

2022/23 UEFA Champions League technical report overview - Open source

Topics: build-up, goal-kicks, technical-report, competition-trends | **Code:** association-football-11v11 | **Type:** competition-technical-report

Context: UEFA Champions League 2022/23 senior men's club competition

Season-specific report overview includes build-up and goal-kick observations; statistics are not universal prescriptions.

Use: Public source link; paraphrase briefly and retain season/competition attribution. Do not copy report text or graphics.

2023/24 UEFA club competition technical report overview - Open source

Topics: compact-defending, pressing, marking-space, women-football, technical-report | **Code:** association-football-11v11 | **Type:** competition-technical-report

Context: UEFA Champions League, Women's Champions League and Europa League 2023/24 senior club competitions

Observer analysis across more than 180 matches includes compact defending, pressing and space-oriented defensive references across competitions.

Use: Public source link; examples remain tied to their competition and season. Do not copy report text or graphics.

2024/25 UEFA club competition technical reports - Open source

Topics: wide-area-defending, goal-kicks, corners, women-football, competition-trends | **Code:** association-football-11v11 | **Type:** competition-technical-report

Context: UEFA senior 2024/25 Champions League, Women's Champions League, Europa League and Conference League

Catalogue of season reports combining coaching perspective, tactical analysis and related practice material across four competitions.

Use: Public source link; preserve competition and season context. Training examples are not match evidence. Do not copy report content.

Paris pressing and mobility in the 2025 Champions League final - Open source

Topics: pressing, mobility, rotations, final-analysis | **Code:** association-football-11v11 | **Type:** match-analysis

Context: UEFA Champions League 2024/25 final, Paris Saint-Germain v Inter

Competition analysis examines coordinated pressing and positional fluidity in one final; treat as a case study, not a causal or universal rule.

Use: Public source link; attribute the specific final. Avoid inferring intent or transferring the outcome to unrelated matches.

The FA Four Corner Model - Open source

Topics: player-development, technical-tactical, physical, psychological, social | **Code:** association-football | **Type:** coach-development-framework

Context: FA player-development guidance with youth examples; adapt to context

Frames player development through connected technical/tactical, physical, psychological and social lenses, considering individual needs and maturation.

Use: Public source link; retain FA attribution. This is a development lens, not a match-event classifier or medical assessment.

Practice design: four-goal games - Open source

Topics: practice-design, small-sided-games, scanning, switching-play, constraints | **Code:** association-football-small-sided-practice | **Type:** practice-design

Context: FA coaching practice example; adapt player numbers, age, ability and objective

Shows how target placement and scoring constraints may encourage scanning and switching while changing resemblance to the full game.

Use: Public source link; practice design is an example. Do not assume training effects transfer directly to match performance.

Match analysis in football: a systematic review - Open source

Topics: match-analysis, operational-definitions, situational-context, methodology | **Code:** association-football | **Type:** peer-reviewed-systematic-review

Context: Review evidence focused on adult men's football; note scope when applying elsewhere

Reviews match-analysis methods and emphasizes operational definitions, standardized categories and situational/interactional context.

Use: Link to PubMed record. Paraphrase and retain population/scope limitations; do not reproduce article text.

Task constraints and tactical behaviours in small-sided football games: a systematic review - Open source

Topics: small-sided-games, task-constraints, tactical-behavior, training-design | **Code:** association-football-small-sided-practice | **Type:** peer-reviewed-systematic-review

Context: Review of football small-sided/conditioned-game studies; samples and task designs vary

Reviews how player numbers, area, rules and other task constraints relate to individual and collective tactical behavior.

Use: DOI link to publisher record. Keep conclusions bounded to included study designs and samples; do not claim automatic match transfer.

Small sided games in soccer: a systematic review - Open source

Topics: small-sided-games, technical-actions, tactical-behavior, physical-demand | **Code:** association-football-small-sided-practice | **Type:** peer-reviewed-systematic-review

Context: Review; included samples and practice designs vary, with many studies on adult men

Synthesizes research on small-sided-game design variables and relationships to technical, tactical and physical outputs.

Use: DOI link to publisher record. Preserve sample and design caveats; do not turn an aggregate review into an individual prescription.

AI methods for football tactical behavior and collective dynamics: a systematic review - Open source

Topics: artificial-intelligence, collective-dynamics, tracking, generalization, methods | **Code:** association-football | **Type:** peer-reviewed-systematic-review

Context: Review with mixed adult/youth and men's/women's eligibility; modalities and designs vary

Maps AI approaches for football tactical patterns, formations, networks and collective behavior, highlighting methodological and sample boundaries.

Use: Open-access article link; paraphrase with citation. Verify publication metadata and evidence scope before relying on a specific result.

Attacking and defensive transitions - Open source

Topics: attacking-transition, defensive-transition, regain, recovery, practice-design | **Code:** association-football-11v11 | **Type:** practice-session

Context: FIFA elite practice session with variable player numbers and overload scenarios

Practice material emphasizes reaction, spatial awareness, support options and movement after possession changes.

Use: A coaching exercise illustrates a practice design, not a universal prescription or observed match fact.

The counter-attack - Open source

Topics: counterattack, transition, forward-runs, rest-defence | **Code:** association-football-11v11 | **Type:** match-analysis

Context: FIFA tournament match-analysis examples; competitions vary within the article

Breaks a counterattack into regain, forward progress, off-ball movement and final action while noting exposure after attacking restarts.

Use: Public source link; source examples do not establish events or intention in a PitchMind match.

Source and interpretation note

This PDF preserves source attribution and context but does not reproduce full articles, images or video. Tactical examples are hypotheses and analytical vocabulary, not evidence about a PitchMind match. The named-coach profiles are not claims of personal experience or imitations. Match facts must come from validated canonical data, and uncertain claims should be narrowed or withheld.

Appendix: PitchMind 18-zone reference

A team-relative location vocabulary for movement, progression, and transition-risk analysis

PITCH-REGIONS-18-V1 6 length bands x left / central / right = 18 numbered tactical regions

OPPONENT GOAL

LEFT	CENTRAL	RIGHT
16 Deep attacking	**17** Deep attacking	**18** Deep attacking
13 High attacking	**14** High attacking	**15** High attacking
10 Attacking midfield	**11** Attacking midfield	**12** Attacking midfield
7 Central midfield	**8** Central midfield	**9** Central midfield
4 Defensive midfield	**5** Defensive midfield	**6** Defensive midfield
1 Deep defensive	**2** Deep defensive	**3** Deep defensive



OWN GOAL

ATTACKING TEAM LEFT TOUCHLINE

ATTACKING TEAM RIGHT TOUCHLINE

How to orient the grid

Each length band reads left-to-right across pitch width as left, central, right; number the bands from the team's own goal toward the opponent goal. Thus 1/2/3 are deep defensive and 16/17/18 are deep attacking. Left/right are from an attacker facing forward and mirror in stadium coordinates when teams change ends.

Key tactical references

Zone 5: defensive-midfield anchor area. **Zone 8:** central-midfield engine room. **Zone 14:** central pocket outside the box. **Zone 17:** central penalty-box reference. Zones 13/15 offer high-wide access; Zones 16/18 refer to deep-wide/byline access.

Conditional transition risk

A defender high at a turnover may leave space behind, but recovery depends on the ball carrier, first pressure, distance, goal-side position, pivot/centre-back cover, far-side narrowing and goalkeeper depth. A zone alone does not prove a mistake or a clear chance.

Zone meaning, boundaries, and evidence rules

Keep PitchMind zone labels separate from formal field markings and measured match evidence

Length band	Left channel	Central channel	Right channel	Typical tactical reference
Deep defensive	1	2	3	Back line / first build-up
Defensive midfield	4	5	6	Outlet / central screen
Central midfield	7	8	9	Circulation / switches
Attacking midfield	10	11	12	Between-line access
High attacking	13	14	15	Width / central creative pocket
Deep attacking	16	17	18	Byline / penalty-box reference

Formal markings are overlays

Zone 17 is the PitchMind central penalty-box reference, not the exact legal penalty-area polygon. Zone 14 is a tactical pocket just outside that area. Goal area, penalty area, penalty mark/arc, goalmouth, corner area and cut-back regions require separate calibrated geometry and versioned definitions. IFAB Law 1 specifies formal field markings; map them only after pitch dimensions and axes are calibrated.

What zone analysis can support

With simultaneous, mapped positions, compare occupation, progression, switches, local overloads, regain locations and box entries. Report source IDs, timestamps, selected-time bound, active players, missingness, and `pitch-regions-18-v1` plus the exact boundary-manifest version. A zone change alone does not establish a line break, danger, intent or advantage.

Operational safeguards

Boundary ownership: zone membership requires explicit, versioned cut-lines; do not invent equal metre or percentage bands from the labels.

Time alignment: never count players from different frames as one simultaneous 2v1/3v2. State frame cadence and missing players.

Transition reasoning: an advanced defender at loss suggests a possible channel. Test ball progression, pressure, recovery route, remaining cover and goalkeeper position before claiming a clear risk or inability to recover.

Unknown calibration: do not assign current candidate MANU-LIV coordinates to the zones until units, axes, origin, dimensions, team directions and boundaries are approved.

Sources and scope

This 18-zone vocabulary is a PitchMind convention, not official FIFA/IFAB numbering. FIFA coaching exercises may use context-specific zones. IFAB Law 1 is authoritative for formal field markings; neither source supplies facts about an individual PitchMind match.

IFAB Law 1: The Field of Play

FIFA: Progressing through the thirds

Zone-by-zone tactical reference

General tactical possibilities for each region; roles and advantages must be established from match evidence

Zone	Region	Tactical interpretation
1	Deep defensive left	Primary left-back defensive territory to contain the opposing wide attacker; can also support a safer first build-up along the touchline.
2	Deep defensive center	Central area directly in front of the team's own goal. Prioritize goal protection, secure clearance, or low-risk build-up involving centre-backs and goalkeeper.
3	Deep defensive right	Primary right-back defensive territory to contain the opposing left-sided attacker and limit delivery from that side.
4	Defensive midfield left	Left-sided transition outlet for a full-back or dropping midfielder to escape central pressure and advance possession.
5	Defensive midfield center	Potential number-six anchor and screen in front of centre-backs; can disrupt central counters and offer a first central outlet.
6	Defensive midfield right	Right-sided transition outlet that can widen the first build-up and stretch the opponent's initial press.
7	Central midfield left	Left central-midfield transition area where a winger may link deeper or a progressive midfielder may carry play forward.
8	Central midfield center	Central circulation area for tempo control, connecting defence to attack, and switching play toward either flank.
9	Central midfield right	Right central-midfield circulation area for retaining possession and changing the angle of attack.
10	Attacking midfield left	Left half-space entry reference where an inverted winger or attacking midfielder may receive between the opposing full-back and centre-back.
11	Attacking midfield center	Central link pocket for receiving on the half-turn and connecting deeper midfield to the final attacking line.
12	Attacking midfield right	Right half-space entry reference where a creative player may receive between the opposing full-back and centre-back and connect into the box.
13	High attacking left	Wide creative area for an overlapping full-back or winger to deliver a cross or cutback; can also support an inside run.
14	High attacking center (Zone 14 reference)	Central creative pocket immediately outside the opposition penalty area. A receiver may draw a centre-back forward and create space behind, depending on defensive cover and movement.
15	High attacking right	Wide creative area for a winger to isolate a defender, cut inside where suitable, or for a full-back to deliver toward the far post.
16	Deep attacking left	Left byline/corner-side area used to pin the defence deep, seek a restart, or deliver a low cross/cutback across goal.
17	Deep attacking center (Zone 17 reference)	Central penalty-box reference where attackers may time runs and defenders protect goal-facing space. The legal penalty-area polygon is a separate calibrated overlay; no goal-frequency claim is implied.
18	Deep attacking right	Right byline/corner-side area used to pin the defence deep, seek a restart, or deliver a low cross/cutback across goal.

Evidence rule: These descriptions are tactical hypotheses attached to a region label. They do not establish a player's role, intent, pressure, tactical error, chance quality or outcome. Use synchronized, calibrated locations and validated event/clock semantics for match-specific claims.